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Open Network for Data Exchange — User Guide

How an agency connects and answers verified questions

ATTACHMENT 2 · USER GUIDE

The Open Network for Data Exchange lets one government agency ask another a verified question about a person or a case — and get a signed answer in seconds — without any agency giving up control of its data. This guide explains, in plain terms, how it works, how an agency connects, and what happens when a question is asked.

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1. The idea in one paragraph

There is no central database. Each agency keeps its own records exactly where they are today. In front of those records sits a small piece of software — a node — that answers authorised questions locally and signs each answer with the agency's own cryptographic key. The network's job is only to carry the question to the right agency and to verify identities, so the receiver can trust that an answer really came from, say, the National Identity Management Commission and no one else. The question and the answer travel; the data never does.

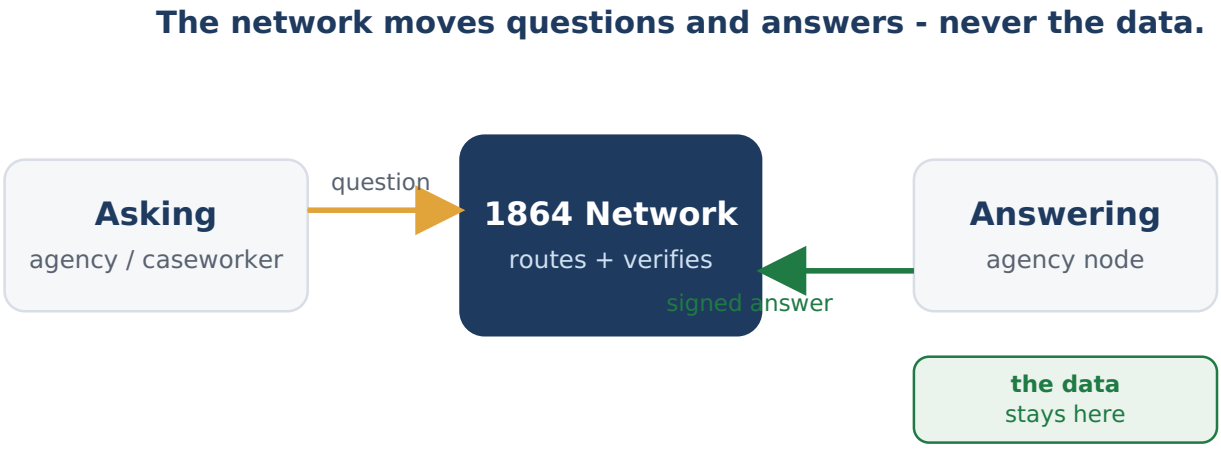


Figure: only a signed answer travels; each agency keeps its records.

2. What actually crosses the wire

Only a signed answer — never the data

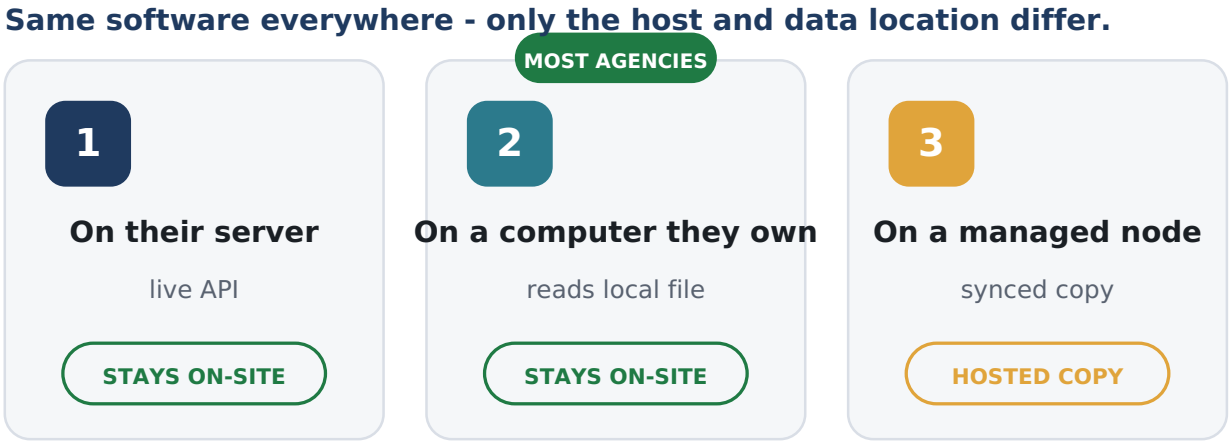
A specific answer to a specific authorised question: "registered", "NIN assigned", "enrolled". Never the file, never the database, never a bulk export. The 1864-operated network routes the question and checks identities; it does not see or store the underlying data.

This is the single most important property of the design, and it is what makes agencies — and data-protection

regulators — comfortable joining. An agency exposes answers to agreed questions, not access to its records.

3. Three ways an agency connects

It is the same software everywhere. What differs is only where it runs and where the data sits. Sensitive data always stays on-site.



Model	For whom	How it works	Data
1 · On their own server	Agencies with a server and an API	The software installs on the server they already run; the connector calls their live system. Answers are always current.	On-site
2 · On a computer they own	Agencies on Excel or files, no server	Installs on an office computer they already have; reads the spreadsheet locally, in the building.	On-site
3 · On a managed node	Where a hosted copy is acceptable	1864 runs the node in the cloud and reads a synced copy. Non-sensitive registers only.	Hosted copy

Every agency has a computer. Where there is no suitable always-on machine, a small pre-configured device can be provided. In every on-site case the only requirement is that the computer stays powered on and connected to the internet during the hours the agency wants to be reachable; where mains power is unreliable, a UPS or inverter covering those hours is recommended.

4. What runs at each agency

One small install sits in front of the data the agency already keeps. It has two parts — a connector and an adapter — and it holds the agency's own signing key, generated on the machine and never shared.

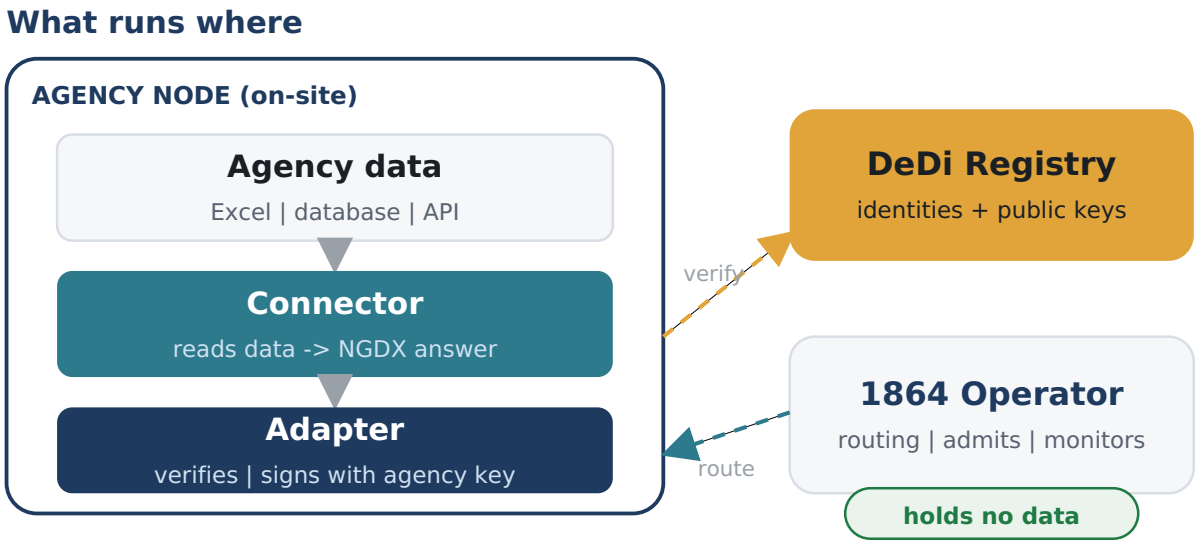


Figure: the node stays with the agency; the operator runs only registry + routing.

The connector
Reads the agency's file or API and returns only the one authorised answer, formatted to Nigeria's data-exchange standard. It never returns the whole record.

The adapter
Checks who is asking (verifying the caller's signature against the shared registry) and signs the answer with the agency's own private key. Nobody else can forge it.

Staff keep working exactly as before — the node runs quietly in the background. For a spreadsheet agency, the only new habit is keeping the file current where the node can read it.

5. What happens when a question is asked

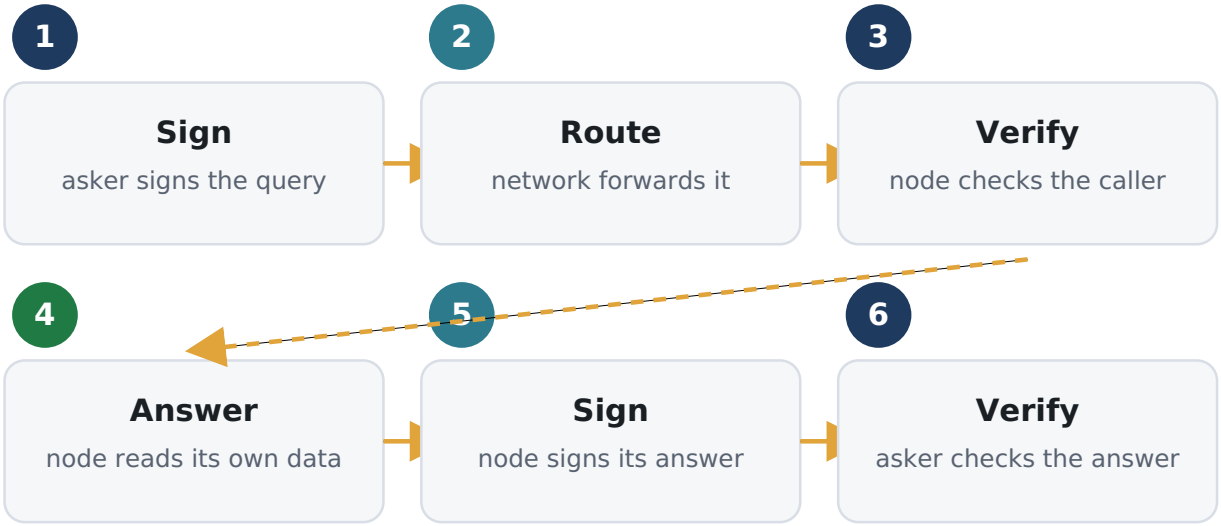


Figure: a verified exchange, from signed question to verified answer.

1. The asking agency’s system signs the question and sends it to the network.
2. The network looks up the asker in the registry and verifies the signature — only known participants can ask.
3. The question reaches the answering agency’s node, which reads its own data and finds the answer.
4. The node signs the answer with the agency’s own key and returns it.
5. The asker verifies that signature against the registry and accepts the answer — confirmed in seconds.

6. A worked example: the birth loop

A caseworker in Kaduna registers a newborn and needs to close a three-agency loop. On the Open Network this runs as three verified questions, each answered by a different agency under its own identity, linked by the National Identification Number.

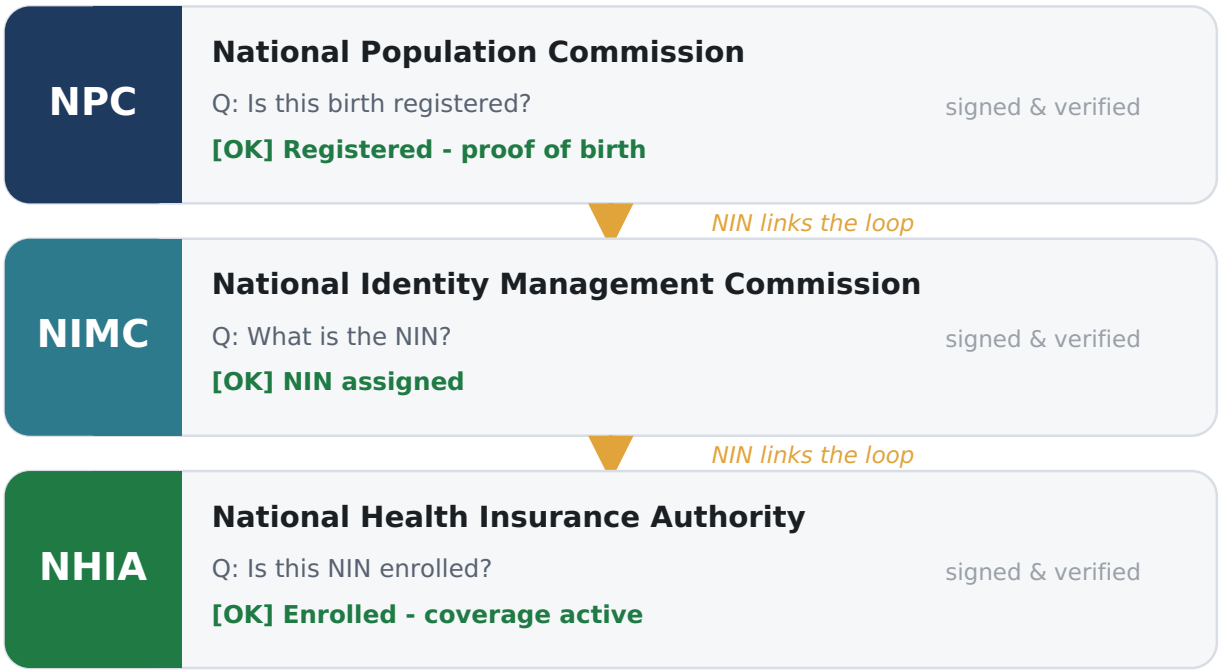


Figure: the birth loop - three agencies, one linked answer, in seconds.

Step	Agency	Question	Signed answer
1	National Population Commission	Is this birth registered?	Registered — proof of birth issued
2	National Identity Management Commission	What is the NIN for this birth?	NIN assigned
3	National Health Insurance Authority	Is this NIN enrolled for coverage?	Enrolled — coverage active

No central database is created; each agency answers from its own records and keeps control of them. In our prototype, all three agencies are registered with real cryptographic identities and each answer verifies successfully — the loop closes end to end.

7. Governance, consent, and trust

- **Decentralised by design.** Admission is through a registry operated by 1864, so states, LGAs, clinics, fintechs, and NGOs can all join on the same terms — with no central hub holding the data.
- **Consent and audit.** The network can enforce which participant may ask which question, and records that a query happened, without seeing its contents.
- **Local key custody.** Each agency’s private key is generated on its own node and never leaves it; the operator cannot sign as an agency or read its data.
- **Open standards.** The protocol is Beckn, the open standard behind India’s national networks, and the field semantics follow Nigeria’s Government Data Exchange standard — so joining means adopting a standard the country has already endorsed, with no vendor lock-in.

Companion documents: the **1864 Prep User Guide** (getting data ready) and the **Technical Documentation** (architecture, protocols, identity, and deployment).